

5781007

B.Tech. DEGREE EXAMINATION,
NOVEMBER/DECEMBER 2018.

Second Semester

(Common to All Branches)

MATHEMATICS – II

(From 2013-14 onwards)

Time : Three hours

Maximum : 75 marks

SECTION A — (10 × 2 = 20 marks)

Answer ALL the questions.

1. Prove that the Eigen vectors corresponding to distinct Eigen values of real symmetric matrix are orthogonal.

2. Determine λ so that

$\lambda(x^2 + y^2 + z^2) + 2xy - 2yz + 2zx$ is positive definite.

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Time : Three hours

Maximum : 75 marks

SECTION A — (10 × 2 = 20 marks)

Answer ALL the questions.

1. Prove that the Eigen vectors corresponding to distinct Eigen values of real symmetric matrix are orthogonal.

2. Determine λ so that

$\lambda(x^2 + y^2 + z^2) + 2xy - 2yz + 2zx$ is positive definite.

3. Prove that gradient of constant is zero.
4. Is the position vector $\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$ irrotational? Justify.
5. Define Laplace Transforms.
6. Verify the final value theorem for $f(t) = 3e^{-t}$.
7. Define inverse Laplace Transforms.
8. Evaluate $L^{-1}\left(\frac{1}{s^2 + 6s + 13}\right)$.
9. State Fourier integral theorem.
10. Prove that $F\left\{\frac{d^n}{dx^n}f(x)\right\} = (-is)^n F(s)$.

SECTION B — (5 × 11 = 55 marks)

Answer ALL the units, choosing ONE question from each Unit.

UNIT I

11. (a) Find the characteristic equation of the matrix $A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & -1 & 4 \\ 3 & 1 & -1 \end{pmatrix}$ and show that A satisfies the equation. Hence evaluation A^{-1} and A^4 .

(a) Find $F_c\{e^{-a^2x^2}\}$ & $F_s\{xe^{-a^2x^2}\}$ and Show that is $e^{x^2/2}$ self reciprocal under Fourier cosine transform and $xe^{-x^2/2}$ is self reciprocal under Fourier sine transform.

(b) Using Fourier integral show that

$$\int_0^{\infty} \frac{(s^2 + 2)\sin(sx)}{s^2 + 4} ds = \frac{\pi}{2} e^{-x} \cos x.$$

(b) Using Cayley – Hamilton theorem, find A^{-1}

if $A = \begin{pmatrix} 1 & 2 & -2 \\ 2 & 5 & -4 \\ 3 & 7 & -5 \end{pmatrix}$. Also verify the theorem.

Or

12. Reduce the quadratic form

$10x_1^2 + 2x_2^2 + 5x_3^2 + 6x_2x_3 - 10x_3x_1 - 4x_1x_2$ to a canonical form by orthogonal reduction. Find a set of values of x_1, x_2, x_3 which will make the form vanish.

UNIT II

13. (a) Find the directional derivative for $f(x, y, z) = 2xy + z^2$ at the point $(1, -1, 3)$ in the direction of the vectors $\vec{i} + 2\vec{j} + 2\vec{k}$. (6)

- (b) Find the values of n such that the vector $r^n \vec{r}$ is both solenoidal and irrotational. (5)

Or

14. Verify Gauss divergence theorem for $\vec{F} = x^2 \vec{i} + z\vec{j} + yz\vec{k}$ over the cube formed by the planes $x = \pm 1, y = \pm 1, z = \pm 1$.

UNIT III

15. (a) Explain first shifting property. (4)
- (b) Verify initial and final value theorems for the function $f(t) = 1 + e^{t(\sin t + \cos t)}$. (7)

Or

16. (a) Prove that $L(tf(t)) = -\frac{d}{ds}F(s)$ if $L(f(t)) = F(s)$. (5)
- (b) Find $L(t^2 e^{-3t} \sin 2t)$. (6)

UNIT IV

17. (a) Evaluate $L^{-1}\left(\frac{1}{(s^2 + a^2)^2}\right)$.
- (b) Find the inverse Linear Transform of following $\log\left(\frac{s+1}{s-1}\right)$.
- (c) Find the inverse Linear Transform $\frac{1}{(s+1)(s^2+4)}$.

Or

18. Solve by the Linear Transform method, equation $\frac{d^2 y}{dt^2} + 2\frac{dy}{dt} + 5y = e^{-t} \sin t, y(0) = 0, y'(0) = 0$.

UNIT V

19. (a) Derive Fourier sine transform.
- (b) State and prove Parseval's identity and prove that $F_C\{xf(x)\} = \frac{dF_s}{ds}$.

Or

UNIT V

19. Find the Fourier transform of $f(x) = \begin{cases} 1-x^2, & |x| < 1 \\ 0 & , |x| > 1 \end{cases}$ and hence evaluate (5)

(a) $\int_0^{\infty} \frac{\sin t - t \cos t}{t^3} dt.$ (3)

(b) $\int_0^{\infty} \left(\frac{x \cos x - \sin x}{x^3} \right) \cos \frac{x}{2} dx.$ (3)

Or

20. Find the Fourier cosine transforms of $f(x) = \begin{cases} 1-x^2, & 0 < x < 1 \\ 0 & , \text{otherwise} \end{cases}$ and hence prove

$$\int_0^{\infty} \left(\frac{\sin x - x \cos x}{x^3} \right) \cos \left(\frac{x}{2} \right) dx = \frac{3\pi}{16}. \quad (11)$$

5781007

B.Tech. DEGREE EXAMINATION, MAY 2019.

Second Semester

MATHEMATICS – II

(Common to All Branches)

(From 2013-14 Onwards)

Time : Three hours

Maximum : 75 marks

SECTION A — (10 × 2 = 20 marks)

Answer ALL the questions.

All questions carry equal marks.

1. Define trace of a matrix.
2. State Cayley-Hamilton theorem.
3. If $\phi = xyz$, then find $\nabla \phi$.
4. What is the angle between the normals to the surface $xy = z^2$ at the points $(1, 4, 2)$ and $(-3, -3, 3)$.
5. Find $L[tf(t)]$.

6. Write down the statement of second shifting theorem.

7. Find $L^{-1}\left[\frac{1}{(s-5)^5}\right]$.

8. Find the Laplace transform of $t^2 + te^{-t} + t^3 \sin 2t$.

9. State Fourier integral theorem.

10. State Convolution theorem.

SECTION B — (5 × 11 = 55 marks)

Answer ALL the units, Choosing ONE full question from each Unit.

UNIT I

11. Find the eigenvalue and eigenvectors of
- $$\begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}. \quad (11)$$

Or

12. Find A^8 for $A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 1 \\ -4 & 4 & 3 \end{bmatrix}$. (11)

UNIT II

13. Prove that $\nabla^2(r^n \vec{r}) = n(n+3)r^{n-2}$.

Or

14. Verify Gauss divergence theorem $\vec{F} = (x^2 - yz)\vec{i} + (y^2 - zx)\vec{j} + (z^2 - xy)\vec{k}$ taken over rectangular parallelepiped $0 \leq x \leq a, 0 \leq y \leq b, 0 \leq z \leq c$. 19.

UNIT III

15. Show that $\int_0^\infty \frac{e^{-t} \sin t}{t} dt = \frac{\pi}{4}$. 20.

Or

16. Verify the initial and final value theorem for the function $f(t) = 1 + e^{-t}(\sin t + \cos t)$. 11

UNIT IV

17. Evaluate $L^{-1}\left[\frac{1}{s^3(s^2-1)}\right]$ by convolution theorem. 11

Or

18. Solve $\frac{d^2x}{dt^2} + 3\frac{dy}{dt} + 2x = e^{-t} \sin t$ given that $x(0) = 1$ and $x'(0) = 1$. 19.

17. Using method of partial fraction find

$$L^{-1} \left[\frac{5s^2 - 15s - 11}{(s+1)(s-2)^3} \right]. \quad (11)$$

Or

18. Using Laplace transform solve $y'' - 3y' + 2y = e^{2t}$,
 $y(0) = -3$ and $y'(0) = 5.$ (11)

19. Find the Fourier Transform for $f(x) = \begin{cases} 1, & |x| < a \\ 0, & |x| > a \end{cases}$.

Hence deduce that $\int_0^{\infty} \left(\frac{\sin t}{t} \right) dt = \frac{\pi}{2}.$ (11)

Or

20. (a) Evaluate $\int_0^{\infty} \frac{dx}{(x^2 + a^2)^2}$ using Parseval's identity of Fourier cosine transform. (6)
- (b) Find Fourier sine transform of $f(x) = xe^{-ax}$,
 $a > 0.$ (5)

5781007

B.Tech. DEGREE EXAMINATION, SEPTEMBER 2020.

Second Semester

Common to All Branches

MATHEMATICS - II

(From 2013 - 14 onwards)

Time : Three hours

Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL the questions.

All questions carry equal marks.

1. Verify Cayley Hamilton theorem for $A = \begin{bmatrix} 1 & 2 \\ 4 & 5 \end{bmatrix}$.
2. The two Eigen values of a matrix $A = \begin{bmatrix} 4 & 6 & 6 \\ 1 & 3 & 2 \\ -1 & -5 & -2 \end{bmatrix}$ are equal to the double of the third Eigen value. Find Eigen value of A^2 .
3. Show that Curl grad $\phi = 0$.
4. If S is a closed surface enclosed is the volume V and $\vec{A} = ax\hat{i} + by\hat{j} + cz\hat{k}$. Show that $\iint_S \vec{A} \cdot \hat{n} ds = (a + b + c)V$.

5. Verify Final value theorem on Laplace transform for $f(t) = ae^{-bt}$, $b > 0$.

6. Does $L\left[\frac{\cos at}{t}\right]$ exist?

7. Find $L^{-1}\left[\frac{1}{(S+2)^2}\right]$.

8. State convolution theorem on Laplace inverse transform.

9. State and prove change of scale property of Fourier Transform.

10. Find Fourier Transform for $f(x) = x$, $|x| \leq 1$.

PART B — (5 × 11 = 55 marks)

Answer ALL the Units, choosing ONE full questions from each Unit.

11. Verify Cayley Hamilton theorem for the matrix

$$A = \begin{bmatrix} 1 & 2 & -1 \\ 2 & -3 & 1 \\ 2 & 4 & -2 \end{bmatrix} \text{ and hence find } A^4. \quad (11)$$

Or

12. Diagonalise the matrix $A = \begin{bmatrix} 2 & 2 & 0 \\ 2 & 1 & 1 \\ -7 & 2 & -3 \end{bmatrix}$. (11)

13. Verify Gauss Divergence theorem for $\vec{F} = 4xz\hat{i} - y^2\hat{j} + yz\hat{k}$ where S is the surface of the cube bounded by the lines $x=0, y=0, z=0, x=2, y=2, z=2$. (11)

Or

14. Verify Stoke's theorem for $\vec{A} = x^2\hat{i} + xy\hat{j}$ taken over the square region bounded by the lines $x=0, y=0, x=2, y=2$. (11)

15. Find the Laplace transform of the Half-wave

$$\text{rectifier function } f(t) = \begin{cases} \sin \omega t, & 0 < t < \frac{\pi}{\omega} \\ 0, & \frac{\pi}{\omega} < t < \frac{2\pi}{\omega} \end{cases} \text{ where}$$

$$f\left(t + \frac{2\pi}{\omega}\right) = f(t). \quad (11)$$

Or

16. (a) Find $L[t^2 e^{-3t}]$. (5)

(b) Find $L\left[\frac{1-e^{-t}}{t}\right]$. (6)

UNIT IV

17. (a) Find $L^{-1}\left[\frac{s}{s^2a^2+b^2}\right]$. (6)

(b) Find $L^{-1}\left[\frac{s+2}{(s^2+4s+5)^2}\right]$. (5)

Or

8. Solve the equation $\frac{d^2y}{dt^2} + 2\frac{dy}{dt} - 3y = \sin t$ given that $y = \frac{dy}{dt} = 0$ when $t = 0$.

UNIT V

9. Find the Fourier transform of $f(x) = \begin{cases} 1-x^2, & \text{if } |x| < 1 \\ 0, & \text{if } |x| > 1 \end{cases}$ hence show that $\int_0^\infty \frac{\sin s - s \cos s}{s^2} \cos \frac{s}{2} ds = \frac{3\pi}{16}$.

Or

10. Find the Fourier transform of $e^{-|x|}$ and deduce that $\int_0^\infty \frac{\cos xt}{1+t^2} dt = \frac{1}{2}\pi e^{-|x|}$.

5781007

B.Tech. DEGREE EXAMINATION, APRIL/MAY 2018.

Second Semester

Mathematics – II

(Common to ALL branches)

(From 2013 – 14 onwards)

Time : Three hours

Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL the questions.

All questions carry equal marks.

1. Define eigen values and eigen vectors.
2. State Cayley-Hamilton theorem.
3. Find $\text{grad } \phi$ if $\phi = xyz$ at $(1, 1, 1)$.
4. State Stoke's theorem.
5. Find $L(t^2 + 2t + 3)$.
6. Find $L(t^2 e^{-3t})$.

7. Find $L^{-1}\left[\frac{s-3}{(s-3)^2+4}\right]$.

8. State convolution theorem.

9. State Fourier integral theorem.

10. Show that $f(x)=1$, $0 < x < \infty$ cannot be represented by a Fourier Integral.

PART B — $(5 \times 11 = 55 \text{ marks})$

Answer ALL the units, Choosing ONE full question from each Unit.

UNIT I

11. Find the eigen values and eigen vector of $\begin{bmatrix} 2 & 2 & 0 \\ 2 & 1 & 1 \\ -7 & 2 & -3 \end{bmatrix}$.

Or

12. Verify Cayley-Hamilton theorem and hence find inverse for $\begin{bmatrix} 1 & 3 & 7 \\ 4 & 2 & 3 \\ 1 & 2 & 1 \end{bmatrix}$.

2

UNIT II

13. (a) Find the directional derivative of $\phi = x^2yz + 4xz^2$ at $(1, 1, 1)$ in the direction of $\vec{i} + \vec{j} - \vec{k}$. (6)

(b) Find the divergence and curl of the vector point function $xy^2\vec{i} + 2x^2yz\vec{j} - 3yz^2\vec{k}$. (5)

Or

14. Verify divergence theorem for

$\vec{F} = (x^2 - yz)\vec{i} + (y^2 - zx)\vec{j} + (z^2 - xy)\vec{k}$ taken over the rectangular parallelopiped $0 \leq x \leq a$, $0 \leq y \leq b$, $0 \leq z \leq c$.

UNIT III

15. (a) State and prove final value theorem. (6)

(b) Find $L\left[\frac{\sin at}{t}\right]$. (5)

Or

16. (a) State and prove initial value theorem. (5)

(b) Evaluate $\int_0^\infty te^{-3t} \cos t \, dt$. (6)

5781007

3

5781007

UNIT IV

5781007

17. Apply convolution theorem to evaluate
 (a) $L^{-1}\left[\frac{s}{(s^2+a^2)^2}\right]$ (b) $L^{-1}\left[\frac{s^2}{(s^2+a^2)(s^2+b^2)}\right]$

Or

18. Work out $\frac{dx}{dt} + 5x - 2y = t$, $\frac{dy}{dt} + 2x + y = 0$ being given $x = y = 0$ when $t = 0$, by the transform method.

UNIT V

19. Find the Fourier transform of
 $f(x) = \begin{cases} 1-x^2 & \text{if } |x| \leq 1 \\ 0 & \text{if } |x| > 1 \end{cases}$
 Hence evaluate

$$\int_0^{\infty} \frac{x \cos x - \sin x}{x^3} \cos \frac{x}{2} dx.$$

Or

20. Using Parseval's identities, prove that

$$(a) \int_0^{\infty} \frac{dt}{(a^2+t^2)(b^2+t^2)} = \frac{\pi}{2ab(a+b)}.$$

$$(b) \int_0^{\infty} \frac{\sin at}{t(a^2+t^2)} dt = \frac{\pi}{2} \cdot \frac{1-e^{-a^2}}{a^2}.$$

B.Tech. DEGREE EXAMINATION,
 NOVEMBER/DECEMBER 2017.

Second Semester

MATHEMATICS — II

(From 2013-2014 onwards)

(Common to All Branches)

Time : Three hours

Maximum : 75 marks

SECTION A — (10 × 2 = 20 marks)

Answer ALL the questions.

All questions carry equal marks.

1. Find the eigen values of the matrix $\begin{pmatrix} 5 & 4 \\ 1 & 2 \end{pmatrix}$.

2. Find the product of the eigen values of

$$\begin{pmatrix} 7 & 2 & 2 \\ -6 & -1 & 2 \\ 6 & 2 & -1 \end{pmatrix}.$$

3. Find $\nabla \phi$, if $\phi = \log(x^2 + y^2 + z^2)$.

4. If $F = 3xyI - y^2J$, evaluate $\int F \cdot dR$, where C is the curve in the xy -plane $y = 2x^2$ from $(0, 0)$ to $(1, 2)$.

5. Show that $L(e^{at}) = \frac{1}{s-a}$.
6. Find $L(\sin 2t \sin 3t)$.
7. Find $L^{-1}\left[\frac{1}{(s-a)^2 + b^2}\right]$.
8. State Convolution theorem.
9. Write down the complex form of the Fourier integral.

10. Find the Fourier sine transform of $\frac{e^{-ax}}{x}$.

SECTION B — (5 × 11 = 55 marks)

Answer ALL the Units, choosing ONE from each Unit.

UNIT I

11. Find the characteristic equation of the matrix $A = \begin{pmatrix} 1 & 1 & 3 \\ 1 & 3 & -3 \\ -2 & -4 & -4 \end{pmatrix}$ and hence find its inverse.

Or

2

5781007

12. Reduce the quadratic form $3x^2 + 5y^2 + 3z^2 - 2yz + 2zx - 2xy$ to the canonical form. Also specify the matrix of transformation.

UNIT II

13. Verify Green's theorem for $\int_C [(xy + y^2)dx + x^2dy]$ where C is bounded by $y = x$ and $y = x^2$.

Or

14. Evaluate $\int_S F \cdot dS$ where $F = 4xI - 2y^2J + z^2K$ and S is the surface bounding the region $x^2 + y^2 = 4$, $z = 0$ and $z = 3$.

UNIT III

15. Find the Laplace transforms of (a) $t \cos at$ (b) $te^{-t} \sin 3t$.

Or

16. Find the inverse transform of $\frac{5s+3}{(s-1)(s^2+2s+5)}$.

3

5781007

UNIT IV

17. Find the inverse Laplace transform of $\frac{s^2}{(s^2 + a^2)(s^2 + b^2)}$ using convolution theorem.

Or

18. Solve $y'' - 4y' + 3y = e^{-t}$, given $y(0) = 1$, $y'(0) = 0$ using Laplace transform.

UNIT V

19. Find the Fourier transform of $f(x) = \begin{cases} 1 - x^2, & \text{if } |x| < 1 \\ 0, & \text{if } |x| > 1. \end{cases}$ Hence prove that

$$\int_0^{\infty} \frac{\sin s - s \cos s}{s^3} \cos \frac{s}{2} ds = \frac{3\pi}{16}.$$

Or

20. Evaluate $\int_0^{\infty} \frac{dx}{(x^2 + a^2)(x^2 + b^2)}$ using Fourier cosine transform.

5781007

B.Tech. DEGREE EXAMINATION, MAY 2016.

Second Semester

Common to All Branches

MATHEMATICS — II

(From 2013-14 batch onwards)

Time : Three hours

Maximum : 75 marks

SECTION A — (10 × 2 = 20 marks)

Answer ALL the questions.

All questions carry equal marks.

1. State Cayley Hamilton theorem.

2. Find the sum and product of Eigen values of the matrix.

$$\begin{bmatrix} 3 & -1 & 1 \\ -1 & 5 & -1 \\ 1 & -1 & 3 \end{bmatrix}.$$

3. If $\phi = x^2 + y^2 + z^2 - 8$ then find $\text{grad } \phi$ at $(2, 0, 2)$.
4. State Stoke's theorem.

5. Find Laplace transform of $t \sin 2t$.
6. Verify initial value theorem for $f(t) = 1 + e^{-t}(\sin t + \cos t)$.
7. Find $L^{-1}\left[\frac{1}{(s-3)^2}\right]$.
8. State the convolution theorem.
9. Write the Fourier transform pair.
10. Find the Fourier sine transform of $\frac{1}{x}$.

SECTION B — (5 × 11 = 55 marks)

Answer ALL questions, choosing ONE from each Unit.

UNIT I

11. Find the eigen values and eigen vectors of $\begin{bmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{bmatrix}$.

Or

12. Diagonalize the matrix $A = \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$ by orthogonal transformation.

UNIT II

13. Verify Gauss divergence theorem for $\vec{F} = 4xz\vec{i} - y^2\vec{j} + yz\vec{k}$ over the cube bounded by $x=0, x=1, y=0, y=1, z=0, z=1$.

Or

14. Verify Stoke's theorem for $\vec{F} = (x^2 - y^2)\vec{i} + 2xy\vec{j}$ in the rectangle region in the xy -plane bounded by the lines $x=0, x=a, y=0$ and $y=b$.

UNIT III

15. Find the Laplace transform of $f(t) = \begin{cases} t & \text{for } 0 < t < a \\ 2a - t & \text{for } a < t < 2a. \end{cases}$ Where $f(t+2a) = f(t)$.

Or

16. (a) Find $L[t^2 e^{-3t} \sin 2t]$.
(b) Evaluate $\int_0^\infty e^{-3t} \cos 4t dt$, using Laplace transform.

18. Evaluate $\iint_S \text{Curl } \vec{F} \cdot \vec{n} \, ds$ where $\vec{F} = (y - z)\vec{i} + yz\vec{j} - xz\vec{k}$ and S is the open surface bounded by the planes $x = 0, x = 1, y = 0, y = 1, z = 0, z = 1$ above the xy plane.

UNIT V

19. X is a normal distribution with mean 16 and standard deviation 3. Find

- (a) $p(x \geq 19)$
 (b) $p(12.5 < x < 19)$
 (c) $p(10 < x < 25)$.

Or

20. Following table gives the data on rainfall and discharge in a certain river. Obtain the line of regression of Y on X .

Rainfall (inches) X :	1.53	1.78	2.60	2.95	3.42
Discharge(1000cc) Y :	33.5	36.3	40.0	45.8	53.5